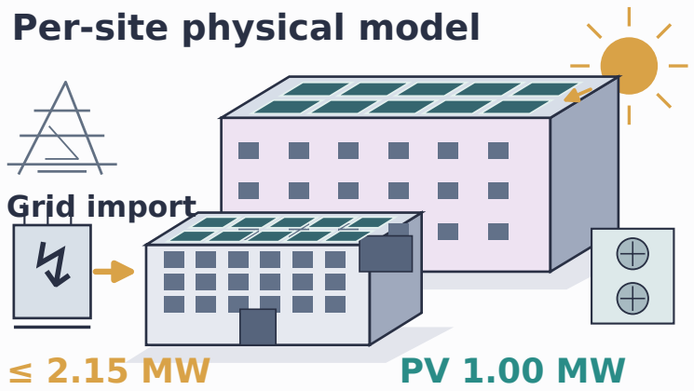
a Data-centre portfolio

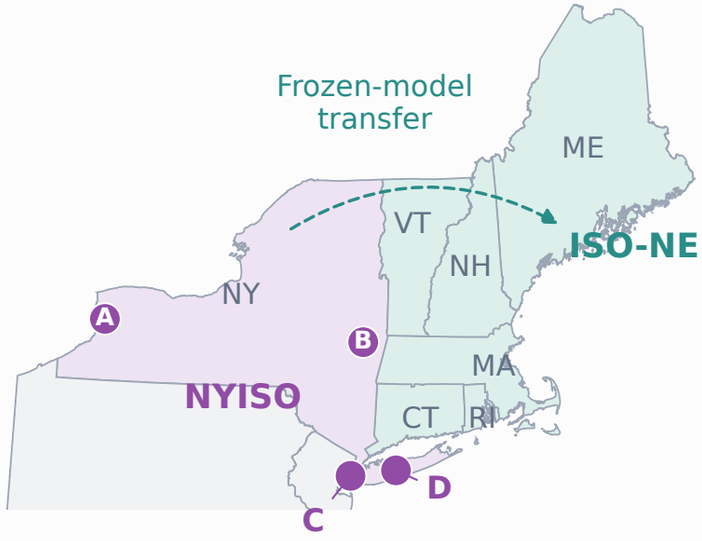
Per-site physical model



IT 1.64 MW + idle 0.23 MW

PUE 1.02–1.09 · zero exports

Configured NYISO locations



A WEST · Buffalo **B CAPITL · Albany**

C N.Y.C. · New York **D LONGIL · Long Island**

Modelled sites; Census state outlines

b Executable-service mechanism

Issued prices + lagged PV / PUE

WorldMoE slot valuation

Frozen features + current queue state

Conserved workload ledger

$$q_{t+1} = q_t + a_t - x_t$$

Unfinished · Released · Executed

24 h service window



Service time moves; work is conserved

Execute one 15 min action

Capacity · deadlines · power balance

Late work $< 10^{-8}$ **IT MWh**

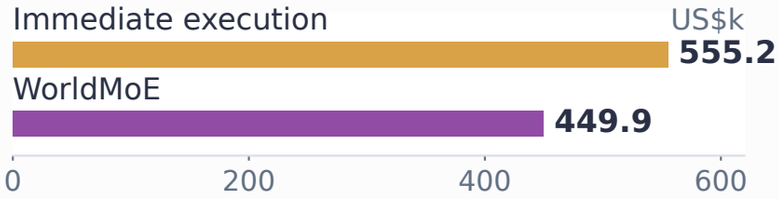
Closing work: 0

c Value and grid response

2025 · 24 h · three refinements

US$105,271

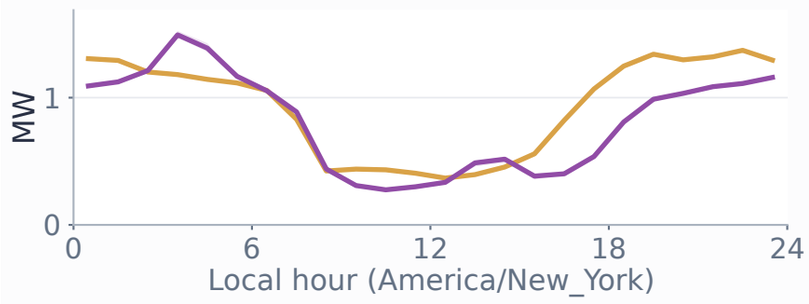
Annual saving · 18.96% of reference bill



79.39% of certified opportunity

Savings ceiling: US$132,605

Full-year mean hourly imports



Highest-price 10% of hours

Imports −23.82% (235.6 MWh)

95th-percentile absolute ramp

1.2432 → 0.7607 MW / 15 min

Annual hourly peak

8.1737 → 8.1742 MW